MSG No. 209.49 $F4321'$ [Type II, cubic]

Table 1: Wyckoff site: 4a, site symmetry: 4321'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120]
2	$[0, \frac{1}{2}, \frac{1}{2}]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,
		121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144]
3	$[\frac{1}{2}, 0, \frac{1}{2}]$	[49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,
		145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168]
4	$[\frac{1}{2}, \frac{1}{2}, 0]$	[73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,
		169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192]

Table 2: Wyckoff site: 4b, site symmetry: 4321'

No.	position	mapping
1	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120]
2	$[\frac{1}{2}, 0, 0]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,
		121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144]
3	$[0, \frac{1}{2}, 0]$	[49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,
		145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168]
4	$[0, 0, \frac{1}{2}]$	[73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,
		169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192]

Table 3: Wyckoff site: 8c, site symmetry: 23.1'

No.	position	mapping
1	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[1,17,18,32,46,47,57,67,72,82,92,93,97,113,114,128,142,143,153,163,168,178,188,189]
2	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[2,7,15,27,28,35,60,62,64,77,78,85,98,103,111,123,124,131,156,158,160,173,174,181]
3	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[3,4,11,26,31,39,53,54,61,84,86,88,99,100,107,122,127,135,149,150,157,180,182,184]
4	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[5,6,13,36,38,40,51,52,59,74,79,87,101,102,109,132,134,136,147,148,155,170,175,183]
5	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[8,22,23,25,41,42,58,68,69,81,91,96,104,118,119,121,137,138,154,164,165,177,187,192]
6	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[9,19,24,34,44,45,49,65,66,80,94,95,105,115,120,130,140,141,145,161,162,176,190,191]
7	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[10,20,21,33,43,48,56,70,71,73,89,90,106,116,117,129,139,144,152,166,167,169,185,186]
8	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[12,14,16,29,30,37,50,55,63,75,76,83,108,110,112,125,126,133,146,151,159,171,172,179]

Table 4: Wyckoff site: 24d, site symmetry: 2.221'

No.	position	mapping
1	$[0, \frac{1}{4}, \frac{1}{4}]$	[1,13,32,38,97,109,128,134]
2	$[0, \frac{3}{4}, \frac{1}{4}]$	[2,10,27,33,98,106,123,129]
3	$[0, \frac{1}{4}, \frac{3}{4}]$	[3,9,26,34,99,105,122,130]
4	$[\frac{1}{4}, \frac{1}{4}, 0]$	[4,18,88,93,100,114,184,189]
5	$[\frac{3}{4}, \frac{1}{4}, 0]$	[5,19,87,95,101,115,183,191]

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[6,20,59,70,102,116,155,166]
7	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[7,17,60,72,103,113,156,168]
8	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[8,14,25,37,104,110,121,133]
9	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[11,22,54,68,107,118,150,164]
10	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[12,24,55,65,108,120,151,161]
11	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[15,23,77,91,111,119,173,187]
12	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[16,21,76,90,112,117,172,186]
13	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[28,42,64,69,124,138,160,165]
14	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[29,43,63,71,125,139,159,167]
15	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[30,44,83,94,126,140,179,190]
16	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[31,41,84,96,127,137,180,192]
17	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[35,46,78,92,131,142,174,188]
18	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[36,48,79,89,132,144,175,185]
19	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[39,47,53,67,135,143,149,163]
20	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[40,45,52,66,136,141,148,162]
21	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[49,61,80,86,145,157,176,182]
22	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[50,58,75,81,146,154,171,177]
23	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[51,57,74,82,147,153,170,178]
24	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[56,62,73,85,152,158,169,181]

Table 5: Wyckoff site: 24e, site symmetry: 4..1'

No.	position	mapping
1	$[x, 0, 0]$	[1,2,3,8,97,98,99,104]
2	$[0, 0, -x]$	[4,16,19,23,100,112,115,119]
3	$[0, 0, x]$	[5,15,18,21,101,111,114,117]
4	$[0, x, 0]$	[6,11,17,24,102,107,113,120]
5	$[0, -x, 0]$	[7,12,20,22,103,108,116,118]
6	$[-x, 0, 0]$	[9,10,13,14,105,106,109,110]
7	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,32,121,122,123,128]
8	$\left[0, \frac{1}{2}, \frac{1}{2} - x\right]$	[28,40,43,47,124,136,139,143]
9	$\left[0, \frac{1}{2}, x + \frac{1}{2}\right]$	[29,39,42,45,125,135,138,141]
10	$\left[0, x + \frac{1}{2}, \frac{1}{2}\right]$	[30,35,41,48,126,131,137,144]
11	$\left[0, \frac{1}{2} - x, \frac{1}{2}\right]$	[31,36,44,46,127,132,140,142]
12	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[33,34,37,38,129,130,133,134]
13	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	[49,50,51,56,145,146,147,152]
14	$\left[\frac{1}{2}, 0, \frac{1}{2} - x\right]$	[52,64,67,71,148,160,163,167]
15	$\left[\frac{1}{2}, 0, x + \frac{1}{2}\right]$	[53,63,66,69,149,159,162,165]
16	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[54,59,65,72,150,155,161,168]
17	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[55,60,68,70,151,156,164,166]
18	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	[57,58,61,62,153,154,157,158]
19	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	[73,74,75,80,169,170,171,176]
20	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[76,88,91,95,172,184,187,191]
21	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[77,87,90,93,173,183,186,189]
22	$\left[\frac{1}{2}, x + \frac{1}{2}, 0\right]$	[78,83,89,96,174,179,185,192]

continued ...

Table 5

No.	position	mapping
23	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[79,84,92,94,175,180,188,190]
24	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[81,82,85,86,177,178,181,182]

Table 6: Wyckoff site: 32f, site symmetry: .3.1'

No.	position	mapping
1	$[x, x, x]$	[1,17,18,97,113,114]
2	$[x, -x, x]$	[2,7,15,98,103,111]
3	$[x, x, -x]$	[3,4,11,99,100,107]
4	$[-x, x, x]$	[5,6,13,101,102,109]
5	$[x, -x, -x]$	[8,22,23,104,118,119]
6	$[-x, x, -x]$	[9,19,24,105,115,120]
7	$[-x, -x, x]$	[10,20,21,106,116,117]
8	$[-x, -x, -x]$	[12,14,16,108,110,112]
9	$\left[x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[25,41,42,121,137,138]
10	$\left[x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[26,31,39,122,127,135]
11	$\left[x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[27,28,35,123,124,131]
12	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[29,30,37,125,126,133]
13	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[32,46,47,128,142,143]
14	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[33,43,48,129,139,144]
15	$\left[-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[34,44,45,130,140,141]
16	$\left[-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[36,38,40,132,134,136]
17	$\left[x+\frac{1}{2}, x, x+\frac{1}{2}\right]$	[49,65,66,145,161,162]
18	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[50,55,63,146,151,159]
19	$\left[x+\frac{1}{2}, x, \frac{1}{2}-x\right]$	[51,52,59,147,148,155]
20	$\left[\frac{1}{2}-x, x, x+\frac{1}{2}\right]$	[53,54,61,149,150,157]
21	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-x\right]$	[56,70,71,152,166,167]
22	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[57,67,72,153,163,168]
23	$\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$	[58,68,69,154,164,165]
24	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-x\right]$	[60,62,64,156,158,160]
25	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x\right]$	[73,89,90,169,185,186]
26	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x\right]$	[74,79,87,170,175,183]
27	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[75,76,83,171,172,179]
28	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x\right]$	[77,78,85,173,174,181]
29	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$	[80,94,95,176,190,191]
30	$\left[\frac{1}{2}-x, x+\frac{1}{2}, -x\right]$	[81,91,96,177,187,192]
31	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[82,92,93,178,188,189]
32	$\left[\frac{1}{2}-x, \frac{1}{2}-x, -x\right]$	[84,86,88,180,182,184]

Table 7: Wyckoff site: 48g, site symmetry: ..21'

No.	position	mapping
1	$[0, y, y]$	[1,13,97,109]
2	$[0, -y, y]$	[2,10,98,106]

continued ...

Table 7

No.	position	mapping
3	$[0,\ y,\ -y]$	[3,9,99,105]
4	$[y,\ y,\ 0]$	[4,18,100,114]
5	$[-y,\ y,\ 0]$	[5,19,101,115]
6	$[-y,\ 0,\ y]$	[6,20,102,116]
7	$[y,\ 0,\ y]$	[7,17,103,113]
8	$[0,\ -y,\ -y]$	[8,14,104,110]
9	$[y,\ 0,\ -y]$	[11,22,107,118]
10	$[-y,\ 0,\ -y]$	[12,24,108,120]
11	$[y,\ -y,\ 0]$	[15,23,111,119]
12	$[-y,\ -y,\ 0]$	[16,21,112,117]
13	$\left[0,\ y+\frac{1}{2},\ y+\frac{1}{2}\right]$	[25,37,121,133]
14	$\left[0,\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	[26,34,122,130]
15	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	[27,33,123,129]
16	$\left[y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[28,42,124,138]
17	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[29,43,125,139]
18	$\left[-y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[30,44,126,140]
19	$\left[y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[31,41,127,137]
20	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$	[32,38,128,134]
21	$\left[y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[35,46,131,142]
22	$\left[-y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[36,48,132,144]
23	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[39,47,135,143]
24	$\left[-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[40,45,136,141]
25	$\left[\frac{1}{2},\ y,\ y+\frac{1}{2}\right]$	[49,61,145,157]
26	$\left[\frac{1}{2},\ -y,\ y+\frac{1}{2}\right]$	[50,58,146,154]
27	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-y\right]$	[51,57,147,153]
28	$\left[y+\frac{1}{2},\ y,\ \frac{1}{2}\right]$	[52,66,148,162]
29	$\left[\frac{1}{2}-y,\ y,\ \frac{1}{2}\right]$	[53,67,149,163]
30	$\left[\frac{1}{2}-y,\ 0,\ y+\frac{1}{2}\right]$	[54,68,150,164]
31	$\left[y+\frac{1}{2},\ 0,\ y+\frac{1}{2}\right]$	[55,65,151,161]
32	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}-y\right]$	[56,62,152,158]
33	$\left[y+\frac{1}{2},\ 0,\ \frac{1}{2}-y\right]$	[59,70,155,166]
34	$\left[\frac{1}{2}-y,\ 0,\ \frac{1}{2}-y\right]$	[60,72,156,168]
35	$\left[y+\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	[63,71,159,167]
36	$\left[\frac{1}{2}-y,\ -y,\ \frac{1}{2}\right]$	[64,69,160,165]
37	$\left[\frac{1}{2},\ y+\frac{1}{2},\ y\right]$	[73,85,169,181]
38	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ y\right]$	[74,82,170,178]
39	$\left[\frac{1}{2},\ y+\frac{1}{2},\ -y\right]$	[75,81,171,177]
40	$\left[y+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[76,90,172,186]
41	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ 0\right]$	[77,91,173,187]
42	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ y\right]$	[78,92,174,188]
43	$\left[y+\frac{1}{2},\ \frac{1}{2},\ y\right]$	[79,89,175,185]
44	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -y\right]$	[80,86,176,182]
45	$\left[y+\frac{1}{2},\ \frac{1}{2},\ -y\right]$	[83,94,179,190]
46	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ -y\right]$	[84,96,180,192]
47	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[87,95,183,191]
48	$\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ 0\right]$	[88,93,184,189]

Table 8: Wyckoff site: `48h`, site symmetry: `..21'`

No.	position	mapping
1	$\left[\frac{1}{2}, y, y\right]$	`[1,13,97,109]`
2	$\left[\frac{1}{2}, -y, y\right]$	`[2,10,98,106]`
3	$\left[\frac{1}{2}, y, -y\right]$	`[3,9,99,105]`
4	$\left[y, y, \frac{1}{2}\right]$	`[4,18,100,114]`
5	$\left[-y, y, \frac{1}{2}\right]$	`[5,19,101,115]`
6	$\left[-y, \frac{1}{2}, y\right]$	`[6,20,102,116]`
7	$\left[y, \frac{1}{2}, y\right]$	`[7,17,103,113]`
8	$\left[\frac{1}{2}, -y, -y\right]$	`[8,14,104,110]`
9	$\left[y, \frac{1}{2}, -y\right]$	`[11,22,107,118]`
10	$\left[-y, \frac{1}{2}, -y\right]$	`[12,24,108,120]`
11	$\left[y, -y, \frac{1}{2}\right]$	`[15,23,111,119]`
12	$\left[-y, -y, \frac{1}{2}\right]$	`[16,21,112,117]`
13	$\left[\frac{1}{2}, y+\frac{1}{2}, y+\frac{1}{2}\right]$	`[25,37,121,133]`
14	$\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$	`[26,34,122,130]`
15	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$	`[27,33,123,129]`
16	$\left[y, y+\frac{1}{2}, 0\right]$	`[28,42,124,138]`
17	$\left[-y, y+\frac{1}{2}, 0\right]$	`[29,43,125,139]`
18	$\left[-y, 0, y+\frac{1}{2}\right]$	`[30,44,126,140]`
19	$\left[y, 0, y+\frac{1}{2}\right]$	`[31,41,127,137]`
20	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-y\right]$	`[32,38,128,134]`
21	$\left[y, 0, \frac{1}{2}-y\right]$	`[35,46,131,142]`
22	$\left[-y, 0, \frac{1}{2}-y\right]$	`[36,48,132,144]`
23	$\left[y, \frac{1}{2}-y, 0\right]$	`[39,47,135,143]`
24	$\left[-y, \frac{1}{2}-y, 0\right]$	`[40,45,136,141]`
25	$\left[0, y, y+\frac{1}{2}\right]$	`[49,61,145,157]`
26	$\left[0, -y, y+\frac{1}{2}\right]$	`[50,58,146,154]`
27	$\left[0, y, \frac{1}{2}-y\right]$	`[51,57,147,153]`
28	$\left[y+\frac{1}{2}, y, 0\right]$	`[52,66,148,162]`
29	$\left[\frac{1}{2}-y, y, 0\right]$	`[53,67,149,163]`
30	$\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$	`[54,68,150,164]`
31	$\left[y+\frac{1}{2}, \frac{1}{2}, y+\frac{1}{2}\right]$	`[55,65,151,161]`
32	$\left[0, -y, \frac{1}{2}-y\right]$	`[56,62,152,158]`
33	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-y\right]$	`[59,70,155,166]`
34	$\left[\frac{1}{2}-y, \frac{1}{2}, \frac{1}{2}-y\right]$	`[60,72,156,168]`
35	$\left[y+\frac{1}{2}, -y, 0\right]$	`[63,71,159,167]`
36	$\left[\frac{1}{2}-y, -y, 0\right]$	`[64,69,160,165]`
37	$\left[0, y+\frac{1}{2}, y\right]$	`[73,85,169,181]`
38	$\left[0, \frac{1}{2}-y, y\right]$	`[74,82,170,178]`
39	$\left[0, y+\frac{1}{2}, -y\right]$	`[75,81,171,177]`
40	$\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$	`[76,90,172,186]`
41	$\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$	`[77,91,173,187]`
42	$\left[\frac{1}{2}-y, 0, y\right]$	`[78,92,174,188]`
43	$\left[y+\frac{1}{2}, 0, y\right]$	`[79,89,175,185]`
44	$\left[0, \frac{1}{2}-y, -y\right]$	`[80,86,176,182]`
45	$\left[y+\frac{1}{2}, 0, -y\right]$	`[83,94,179,190]`
46	$\left[\frac{1}{2}-y, 0, -y\right]$	`[84,96,180,192]`

continued ...

Table 8

No.	position	mapping
47	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[87,95,183,191]
48	$\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[88,93,184,189]

Table 9: Wyckoff site: 48i, site symmetry: 2..1'

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,32,97,128]
2	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[2,27,98,123]
3	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[3,26,99,122]
4	$\left[\frac{1}{4},\ \frac{1}{4},\ -x\right]$	[4,88,100,184]
5	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	[5,87,101,183]
6	$\left[\frac{3}{4},\ x,\ \frac{1}{4}\right]$	[6,59,102,155]
7	$\left[\frac{1}{4},\ -x,\ \frac{1}{4}\right]$	[7,60,103,156]
8	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[8,25,104,121]
9	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[9,34,105,130]
10	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[10,33,106,129]
11	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	[11,54,107,150]
12	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	[12,55,108,151]
13	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[13,38,109,134]
14	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[14,37,110,133]
15	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	[15,77,111,173]
16	$\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$	[16,76,112,172]
17	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	[17,72,113,168]
18	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	[18,93,114,189]
19	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	[19,95,115,191]
20	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[20,70,116,166]
21	$\left[\frac{3}{4},\ \frac{3}{4},\ x\right]$	[21,90,117,186]
22	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	[22,68,118,164]
23	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	[23,91,119,187]
24	$\left[\frac{3}{4},\ x,\ \frac{3}{4}\right]$	[24,65,120,161]
25	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[28,64,124,160]
26	$\left[\frac{3}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[29,63,125,159]
27	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[30,83,126,179]
28	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[31,84,127,180]
29	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[35,78,131,174]
30	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[36,79,132,175]
31	$\left[\frac{1}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[39,53,135,149]
32	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[40,52,136,148]
33	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[41,96,137,192]
34	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[42,69,138,165]
35	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[43,71,139,167]
36	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[44,94,140,190]
37	$\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[45,66,141,162]
38	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[46,92,142,188]
39	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[47,67,143,163]

continued ...

Table 9

No.	position	mapping
40	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[48,89,144,185]
41	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[49,80,145,176]
42	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[50,75,146,171]
43	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[51,74,147,170]
44	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[56,73,152,169]
45	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[57,82,153,178]
46	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{3}{4}\right]$	[58,81,154,177]
47	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[61,86,157,182]
48	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[62,85,158,181]

Table 10: Wyckoff site: 96j, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,97]
2	$[x, -z, y]$	[2,98]
3	$[x, z, -y]$	[3,99]
4	$[z, y, -x]$	[4,100]
5	$[-z, y, x]$	[5,101]
6	$[-y, x, z]$	[6,102]
7	$[y, -x, z]$	[7,103]
8	$[x, -y, -z]$	[8,104]
9	$[-x, y, -z]$	[9,105]
10	$[-x, -y, z]$	[10,106]
11	$[y, x, -z]$	[11,107]
12	$[-y, -x, -z]$	[12,108]
13	$[-x, z, y]$	[13,109]
14	$[-x, -z, -y]$	[14,110]
15	$[z, -y, x]$	[15,111]
16	$[-z, -y, -x]$	[16,112]
17	$[z, x, y]$	[17,113]
18	$[y, z, x]$	[18,114]
19	$[-y, z, -x]$	[19,115]
20	$[-z, -x, y]$	[20,116]
21	$[-y, -z, x]$	[21,117]
22	$[z, -x, -y]$	[22,118]
23	$[y, -z, -x]$	[23,119]
24	$[-z, x, -y]$	[24,120]
25	$\left[x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[25,121]
26	$\left[x, \frac{1}{2}-z, y+\frac{1}{2}\right]$	[26,122]
27	$\left[x, z+\frac{1}{2}, \frac{1}{2}-y\right]$	[27,123]
28	$\left[z, y+\frac{1}{2}, \frac{1}{2}-x\right]$	[28,124]
29	$\left[-z, y+\frac{1}{2}, x+\frac{1}{2}\right]$	[29,125]
30	$\left[-y, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[30,126]
31	$\left[y, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[31,127]
32	$\left[x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[32,128]

continued ...

Table 10

No.	position	mapping
33	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[33,129]
34	$\left[-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[34,130]
35	$\left[y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[35,131]
36	$\left[-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[36,132]
37	$\left[-x,\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[37,133]
38	$\left[-x,\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[38,134]
39	$\left[z,\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	[39,135]
40	$\left[-z,\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	[40,136]
41	$\left[z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[41,137]
42	$\left[y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[42,138]
43	$\left[-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[43,139]
44	$\left[-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[44,140]
45	$\left[-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[45,141]
46	$\left[z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[46,142]
47	$\left[y,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[47,143]
48	$\left[-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[48,144]
49	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[49,145]
50	$\left[x+\frac{1}{2},\, -z,\, y+\frac{1}{2}\right]$	[50,146]
51	$\left[x+\frac{1}{2},\, z,\, \frac{1}{2}-y\right]$	[51,147]
52	$\left[z+\frac{1}{2},\, y,\, \frac{1}{2}-x\right]$	[52,148]
53	$\left[\frac{1}{2}-z,\, y,\, x+\frac{1}{2}\right]$	[53,149]
54	$\left[\frac{1}{2}-y,\, x,\, z+\frac{1}{2}\right]$	[54,150]
55	$\left[y+\frac{1}{2},\, -x,\, z+\frac{1}{2}\right]$	[55,151]
56	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	[56,152]
57	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[57,153]
58	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[58,154]
59	$\left[y+\frac{1}{2},\, x,\, \frac{1}{2}-z\right]$	[59,155]
60	$\left[\frac{1}{2}-y,\, -x,\, \frac{1}{2}-z\right]$	[60,156]
61	$\left[\frac{1}{2}-x,\, z,\, y+\frac{1}{2}\right]$	[61,157]
62	$\left[\frac{1}{2}-x,\, -z,\, \frac{1}{2}-y\right]$	[62,158]
63	$\left[z+\frac{1}{2},\, -y,\, x+\frac{1}{2}\right]$	[63,159]
64	$\left[\frac{1}{2}-z,\, -y,\, \frac{1}{2}-x\right]$	[64,160]
65	$\left[z+\frac{1}{2},\, x,\, y+\frac{1}{2}\right]$	[65,161]
66	$\left[y+\frac{1}{2},\, z,\, x+\frac{1}{2}\right]$	[66,162]
67	$\left[\frac{1}{2}-y,\, z,\, \frac{1}{2}-x\right]$	[67,163]
68	$\left[\frac{1}{2}-z,\, -x,\, y+\frac{1}{2}\right]$	[68,164]
69	$\left[\frac{1}{2}-y,\, -z,\, x+\frac{1}{2}\right]$	[69,165]
70	$\left[z+\frac{1}{2},\, -x,\, \frac{1}{2}-y\right]$	[70,166]
71	$\left[y+\frac{1}{2},\, -z,\, \frac{1}{2}-x\right]$	[71,167]
72	$\left[\frac{1}{2}-z,\, x,\, \frac{1}{2}-y\right]$	[72,168]
73	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[73,169]
74	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, y\right]$	[74,170]
75	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, -y\right]$	[75,171]
76	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, -x\right]$	[76,172]
77	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, x\right]$	[77,173]
78	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[78,174]
79	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[79,175]

continued ...

Table 10

No.	position	mapping
80	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[80,176]
81	$\left[\frac{1}{2}-x, y+\frac{1}{2}, -z\right]$	[81,177]
82	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	[82,178]
83	$\left[y+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	[83,179]
84	$\left[\frac{1}{2}-y, \frac{1}{2}-x, -z\right]$	[84,180]
85	$\left[\frac{1}{2}-x, z+\frac{1}{2}, y\right]$	[85,181]
86	$\left[\frac{1}{2}-x, \frac{1}{2}-z, -y\right]$	[86,182]
87	$\left[z+\frac{1}{2}, \frac{1}{2}-y, x\right]$	[87,183]
88	$\left[\frac{1}{2}-z, \frac{1}{2}-y, -x\right]$	[88,184]
89	$\left[z+\frac{1}{2}, x+\frac{1}{2}, y\right]$	[89,185]
90	$\left[y+\frac{1}{2}, z+\frac{1}{2}, x\right]$	[90,186]
91	$\left[\frac{1}{2}-y, z+\frac{1}{2}, -x\right]$	[91,187]
92	$\left[\frac{1}{2}-z, \frac{1}{2}-x, y\right]$	[92,188]
93	$\left[\frac{1}{2}-y, \frac{1}{2}-z, x\right]$	[93,189]
94	$\left[z+\frac{1}{2}, \frac{1}{2}-x, -y\right]$	[94,190]
95	$\left[y+\frac{1}{2}, \frac{1}{2}-z, -x\right]$	[95,191]
96	$\left[\frac{1}{2}-z, x+\frac{1}{2}, -y\right]$	[96,192]